

MAXREFDES61# Code Documentation

V01.00

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Chapter 1

Main Page

1.1 Introduction

This is the code documentation for the MAXREFDES61# subsystem reference design.

The Files page contains the File List page and the Globals page.

The Globals page contains the Functions, Variables, and Macros sub-pages.

Chapter 2

Data Structure Index

2.1 Data Structures

Here are the data structures with brief descriptions:

__FILE	7
------------------------	---

Chapter 3

File Index

3.1 File List

Here is a list of all files with brief descriptions:

src/ gpio.c	9
src/ i2c.c	10
src/ main.c	11
src/ maximDeviceSpecificUtilities.c	13
src/ maximDeviceSpecificUtilities.h	16
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Chapter 4

Data Structure Documentation

4.1 `__FILE` Struct Reference

Data Fields

- int [handle](#)

4.1.1 Detailed Description

Definition at line 28 of file `retarget.c`.

4.1.2 Field Documentation

4.1.2.1 int handle

Definition at line 28 of file `retarget.c`.

The documentation for this struct was generated from the following file:

- `src/retarget.c`

Chapter 5

File Documentation

5.1 src/gpio.c File Reference

```
#include "gpio.h"
```

Functions

- void [MX_GPIO_Init](#) (void)

5.1.1 Function Documentation

5.1.1.1 void MX_GPIO_Init (void)

File Name : [gpio.c](#) Date : 06/06/2014 14:31:14 Description : This file provides code for the configuration of all used GPIO pins.

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CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE. Configure pins as Analog Input Output EVENT_OUT EXTI Free pins are configured automatically as Analog (this feature is enabled through the Code Generation settings)

Definition at line 58 of file gpio.c.

5.2 src/i2c.c File Reference

```
#include "i2c.h"
#include "gpio.h"
```

Functions

- void [MX_I2C1_Init](#) (void)
- void [HAL_I2C_MspInit](#) (I2C_HandleTypeDef *hi2c)
- void [HAL_I2C_MspDeInit](#) (I2C_HandleTypeDef *hi2c)

Variables

- I2C_HandleTypeDef [hi2c1](#)

5.2.1 Function Documentation

5.2.1.1 void HAL_I2C_MspDeInit (I2C_HandleTypeDef * hi2c)

I2C1 GPIO Configuration PB6 ----> I2C1_SCL PB7 ----> I2C1_SDA

Definition at line 87 of file i2c.c.

5.2.1.2 void HAL_I2C_MspInit (I2C_HandleTypeDef * hi2c)

I2C1 GPIO Configuration PB6 ----> I2C1_SCL PB7 ----> I2C1_SDA

Definition at line 64 of file i2c.c.

5.2.1.3 void MX_I2C1_Init (void)

Definition at line 48 of file i2c.c.

5.2.2 Variable Documentation

5.2.2.1 I2C_HandleTypeDef hi2c1

File Name : [I2C.c](#) Date : 06/06/2014 14:31:14 Description : This file provides code for the configuration of the I2C instances.

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Definition at line 45 of file `i2c.c`.

5.3 *src/main.c* File Reference

```
#include "stm32f4xx_hal.h"
#include "i2c.h"
#include "spi.h"
#include "tim.h"
#include "usart.h"
#include "gpio.h"
#include "maximDeviceSpecificUtilities.h"
#include "stdio.h"
#include "stdlib.h"
#include "string.h"
#include "menu.h"
#include "utilities.h"
```

Macros

- `#define MAJOR_REVISION 01`
- `#define MINOR_REVISION 00`

Functions

- `uint16_t g_auADCData[65536] __attribute__((at(0x20000000)))`
- `int main(void)`

5.3.1 Detailed Description

```
Project: maxrefdes61#
Filename: main.c
Description: This module contains the Main application for the
             maxrefdes61 customized firmware
```

Revision History:

6/11/2014 Rev 01.00 GL Initial release.

This code follows the following naming conventions:

char chPmodValue

char (array) sPmodString[16]

float fPmodValue

int nPmodValue

int (array) anPmodValue[16]

uint16_t uPmodValue

uint16_t (array) auPmodValue[16]

uint8_t uchPmodValue

uint8_t (array) auchPmodBuffer[16]

unsigned int unPmodValue

int * punPmodValue

Definition in file [main.c](#).

5.3.2 Macro Definition Documentation

5.3.2.1 #define MAJOR_REVISION 01

Definition at line 94 of file main.c.

5.3.2.2 #define MINOR_REVISION 00

Definition at line 95 of file main.c.

5.3.3 Function Documentation

5.3.3.1 uint16_t g_auADCDData [65536] __attribute__ ((at(0x20000000)))

5.3.3.2 int main (void)

Definition at line 104 of file main.c.

5.4 src/maximDeviceSpecificUtilities.c File Reference

```
#include "maximDeviceSpecificUtilities.h"
#include "stm32f4xx_hal.h"
#include "tim.h"
#include "usart.h"
#include "spi.h"
#include "stdio.h"
#include "string.h"
#include "stm32f4xx_hal_dma.h"
#include "utilities.h"
```

Functions

- uint32_t [StartSampling](#) (uint8_t uchADCChannel, uint32_t unSampleSize, int nSampleRate, uint16_t *auSamples)
Receive a block of samples at a constant rate.
- void [HAL_TIM_PeriodElapsedCallback](#) (TIM_HandleTypeDef *htim)
- void [ContinuousSampling](#) (uint8_t uchADCChannel)
Continuously reads the ADC and displays the data on the Terminal.
- void [ADCInit](#) (void)
Configure ADC input range for each input channel.

Variables

- uint16_t [g_auWriteBuffer](#) [2]
- uint16_t [g_auReadBuffer](#) [2]
- char [sTerminalMessage](#) [100]
- uint16_t * [g_auSamples](#)
- uint32_t [g_unSampleSize](#) =65535
- uint16_t [g_auCount](#) =0
- uint8_t [g_uchReadADCStop](#)

5.4.1 Detailed Description

```
Project: MAXREFDES61#
Filename: maximDeviceSpecificUtilities.c
Description: This module is an embedded controller driver for the MAXREFDES61#.
             It contains high level functions:
             StartSampling(...)
             ContinuousSampling()
             ADCInit()
```

This driver can be dropped into a user's application as a starting point for development of an end application

Revision History:

06-12-14 Rev 01.00 GL Initial release.

This code follows the following naming conventions:

```

char chPmodValue
char (array) sPmodString[16]
float fPmodValue
int nPmodValue
int (array) anPmodValue[16]
uint16_t uPmodValue
uint16_t (array) auPmodValue[16]
uint8_t uchPmodValue
uint8_t (array) auchPmodBuffer[16]
unsigned int unPmodValue
int * punPmodValue

```

Definition in file [maximDeviceSpecificUtilities.c](#).

5.4.2 Function Documentation

5.4.2.1 void ADCInit (void)

Configure ADC input range for each input channel.

Details

This function configures the ADC inputs
Commented examples are provided to easily change the input range of each channel.

Parameters

<i>None</i>	
-------------	--

Return values

<i>None</i>	
-------------	--

Definition at line 220 of file maximDeviceSpecificUtilities.c.

5.4.2.2 void ContinuousSampling (uint8_t uchADCChannel)

Continuously reads the ADC and displays the data on the Terminal.

Details

This function reads the ADC every half of a second and displays the data to the Terminal. Press the ESC key to return to the main menu.

Parameters

<i>in</i>	<i>uchADCChannel</i>	- Channel number
-----------	----------------------	------------------

Return values

<i>None</i>

Definition at line 177 of file maximDeviceSpecificUtilities.c.

5.4.2.3 void HAL_TIM_PeriodElapsedCallback (TIM_HandleTypeDef * *htim*)

Definition at line 147 of file maximDeviceSpecificUtilities.c.

5.4.2.4 uint32_t StartSampling (uint8_t *uchADCChannel*, uint32_t *unSampleSize*, int *nSampleRate*, uint16_t * *auSamples*)

Receive a block of samples at a constant rate.

Details

This function is used to receive a block of samples at a constant sampling rate. The size of the block is defined in uint32_t *unSampleSize*. The sampling rate is defined in int *nSampleRate*. The Sampled data will be stored in an array **auSamples*
50ksps, 20ksps, 10ksps, and 1ksps rates controlled by the timer

Parameters

in	<i>uchADCChannel</i>	- Channel number
in	<i>unSampleSize</i>	- Sample size
in	<i>nSampleRate</i>	- Sample rate
out	* <i>auSamples</i>	- The array to store the data samples

Return values

<i>uint32_t</i>	- Number of samples received
-----------------	------------------------------

Definition at line 83 of file maximDeviceSpecificUtilities.c.

5.4.3 Variable Documentation**5.4.3.1 uint16_t g_auCount =0**

Definition at line 80 of file maximDeviceSpecificUtilities.c.

5.4.3.2 uint16_t g_auReadBuffer[2]

Definition at line 76 of file maximDeviceSpecificUtilities.c.

5.4.3.3 uint16_t* g_auSamples

Definition at line 78 of file maximDeviceSpecificUtilities.c.

5.4.3.4 uint16_t g_auWriteBuffer[2]

Definition at line 76 of file maximDeviceSpecificUtilities.c.

5.4.3.5 uint8_t g_uchReadADCStop

Definition at line 81 of file maximDeviceSpecificUtilities.c.

5.4.3.6 uint32_t g_unSampleSize =65535

Definition at line 79 of file maximDeviceSpecificUtilities.c.

5.4.3.7 char sTerminalMessage[100]

Definition at line 77 of file maximDeviceSpecificUtilities.c.

5.5 src/maximDeviceSpecificUtilities.h File Reference

```
#include "stm32f4xx_hal.h"
```

Functions

- uint32_t [StartSampling](#) (uint8_t uchADCChannel, uint32_t unSampleSize, int nSampleRate, uint16_t *auSamples)
Receive a block of samples at a constant rate.
- void [ContinuousSampling](#) (uint8_t uchADCChannel)
Continuously reads the ADC and displays the data on the Terminal.
- void [ADCInit](#) (void)
Configure ADC input range for each input channel.

5.5.1 Detailed Description

```
Project: MAXREFDES61#
Filename: maximDeviceSpecificUtilities.h
Description: This module is an embedded controller driver for the MAXREFDES61#.
             It contains high level functions:
             StartSampling(...)
             ContinuousSampling()
             ADCInit()

             This driver can be dropped into a user's application as a starting
             point for development of an end application
```

Revision History:

06-12-14 Rev 01.00 GL Initial release.

This code follows the following naming conventions:

char chPmodValue


```

char (array) sPmodString[16]
float fPmodValue
int nPmodValue
int (array) anPmodValue[16]
uint16_t uPmodValue
uint16_t (array) auPmodValue[16]
uint8_t uchPmodValue
uint8_t (array) auchPmodBuffer[16]
unsigned int unPmodValue
int * punPmodValue

```

Definition in file [maximDeviceSpecificUtilities.h](#).

5.5.2 Function Documentation

5.5.2.1 void ADCInit (void)

Configure ADC input range for each input channel.

Details

This function configures the ADC inputs

Commented examples are provided to easily change the input range of each channel.

Parameters

<i>None</i>	
-------------	--

Return values

<i>None</i>	
-------------	--

Definition at line 220 of file maximDeviceSpecificUtilities.c.

5.5.2.2 void ContinuousSampling (uint8_t uchADCChannel)

Continuously reads the ADC and displays the data on the Terminal.

Details

This function reads the ADC every half of a second and displays the data to the Terminal. Press the ESC key to return to the main menu.

Parameters

<i>in</i>	<i>uchADCChannel</i>	- Channel number
-----------	----------------------	------------------

Return values

<i>None</i>

Definition at line 177 of file maximDeviceSpecificUtilities.c.

5.5.2.3 uint32_t StartSampling (uint8_t uchADCChannel, uint32_t unSampleSize, int nSampleRate, uint16_t * auSamples)

Receive a block of samples at a constant rate.

Details

This function is used to receive a block of samples at a constant sampling rate. The size of the block is defined in uint32_t unSampleSize. The sampling rate is defined in int nSampleRate. The Sampled data will be stored in an array *auSamples
50ksps, 20ksps, 10ksps, and 1ksps rates controlled by the timer

Parameters

in	<i>uchADCChannel</i>	- Channel number
in	<i>unSampleSize</i>	- Sample size
in	<i>nSampleRate</i>	- Sample rate
out	<i>*auSamples</i>	- The array to store the data samples

Return values

<i>uint32_t</i>	- Number of samples received
-----------------	------------------------------

Definition at line 83 of file maximDeviceSpecificUtilities.c.

5.6 src/menu.c File Reference

```
#include "stdio.h"
#include "menu.h"
```

Functions

- void [menu_cls](#) ()
Function to clear the screen via Hyperterminal.
- void [menu_print_maxim_banner](#) ()
Print standard Maxim banner at top of Hyperterminal screen.
- void [menu_print_maxim_banner_big](#) ()
Print large Maxim banner at top of Hyperterminal screen.
- void [menu_print_prompt](#) ()
Print a standard prompt for keyboard input ">".
- void [menu_print_line](#) ()

Print one line of dashes across the screen via Hyperterminal.

- `uint8_t menu_retrieve_keypress (uint32_t nUartAddress)`

Get a single keypress via Hyperterminal.

- `void menu_print_main_menu ()`

Print the main menu listing choice of module to test.

- `void menu_print_channel_menu ()`
- `void menu_print_sample_rate_menu ()`

Print a menu listing sample speed choices.

- `void menu_print_sample_size_menu ()`

Print a menu listing sample size choices.

5.6.1 Detailed Description

```
Project: MAXREFDES61#
Filename: menu.c
Description: This module contains all the functions used to
             generate the menus and menu options used to run the
             MAXREFDES61# example firmware.
```

Revision History:

06-13-14 Rev 01.00 GL Initial Release

This code follows the following naming conventions:

`char chPmodValue`

`char (array) sPmodString[16]`

`float fPmodValue`

`int nPmodValue`

`int (array) anPmodValue[16]`

`uint16_t uPmodValue`

`uint16_t (array) auPmodValue[16]`

`uint8_t uchPmodValue`

`uint8_t (array) auchPmodBuffer[16]`

`unsigned int unPmodValue`

`int * punPmodValue`

Definition in file [menu.c](#).

5.6.2 Function Documentation

5.6.2.1 `void menu_cls ( void )`

Function to clear the screen via Hyperterminal.

Parameters

<i>None</i>	
-------------	--

Return values

<i>None</i>	
-------------	--

Definition at line 64 of file menu.c.

5.6.2.2 void menu_print_channel_menu (void)

Definition at line 220 of file menu.c.

5.6.2.3 void menu_print_line ()

Print one line of dashes across the screen via Hyperterminal.

Parameters

<i>None</i>	
-------------	--

Return values

<i>None</i>	
-------------	--

Definition at line 149 of file menu.c.

5.6.2.4 void menu_print_main_menu (void)

Print the main menu listing choice of module to test.

Details.**Return values**

<i>None</i>	
-------------	--

Definition at line 203 of file menu.c.

5.6.2.5 void menu_print_maxim_banner (void)

Print standard Maxim banner at top of Hyperterminal screen.

Parameters

<i>None</i>	
-------------	--

Return values

<i>None</i>	
-------------	--

Definition at line 78 of file menu.c.

5.6.2.6 void menu_print_maxim_banner_big (void)

Print large Maxim banner at top of Hyperterminal screen.

Parameters

<i>None</i>	
-------------	--

Return values

<i>None</i>	
-------------	--

Definition at line 102 of file menu.c.

5.6.2.7 void menu_print_prompt (void)

Print a standard prompt for keyboard input " > ".

Parameters

<i>None</i>	
-------------	--

Return values

<i>None</i>	
-------------	--

Definition at line 135 of file menu.c.

5.6.2.8 void menu_print_sample_rate_menu (void)

Print a menu listing sample speed choices.

Details**Return values**

<i>None</i>	
-------------	--

Definition at line 233 of file menu.c.

5.6.2.9 void menu_print_sample_size_menu (void)

Print a menu listing sample size choices.

Details

Return values

<i>None</i>	
-------------	--

Definition at line 252 of file menu.c.

5.6.2.10 uint8_t menu_retrieve_keypress (uint32_t nUartAddress)

Get a single keypress via Hyperterminal.

Details

Returns ascii character corresponding to keypress

Parameters

<i>in</i>	<i>nUartAddress</i>	- address of the UART peripheral in MicroBlaze memory map
-----------	---------------------	---

Return values

<i>Character, partially</i>	decoded.
-----------------------------	----------

Definition at line 162 of file menu.c.

5.7 src/menu.h File Reference

```
#include "stm32f4xx_hal.h"
#include "stdio.h"
```

Macros

- #define **MAIN_MENU** 0
 // Menu state machine state
- #define **WAIT_KEYPRESS** 1
 // Menu state machine state
- #define **MENU_SAMPLE_SPEED** 2
 // Menu state machine state
- #define **MENU_SAMPLE_SIZE** 3
 // Menu state machine state
- #define **START_CONTINUOUS_SAMPLING** 4
 // Menu state machine state
- #define **START_BLOCK_SAMPLING** 5
 // Menu state machine state
- #define **MENU_CHANNEL** 6
- #define **BASE_FUNCTION_STATE** 10
 // Menu state machine state
- #define **KEYPRESS_ARROW_UP** 240

- *Assign up-arrow an extended ascii code which won't be used elsewhere.*
 • #define `KEYPRESS_ARROW_DOWN` 241
- *Assign up-arrow an extended ascii code which won't be used elsewhere.*
 • #define `KEYPRESS_ARROW_LEFT` 242
- *Assign up-arrow an extended ascii code which won't be used elsewhere.*
 • #define `KEYPRESS_ARROW_RIGHT` 243
- *Assign up-arrow an extended ascii code which won't be used elsewhere.*
 • #define `KEYPRESS_END` 244
- *Assign up-arrow an extended ascii code which won't be used elsewhere.*

Functions

- void `menu_cls` (void)
Function to clear the screen via Hyperterminal.
- void `menu_print_maxim_banner` (void)
Print standard Maxim banner at top of Hyperterminal screen.
- void `menu_print_maxim_banner_big` (void)
Print large Maxim banner at top of Hyperterminal screen.
- void `menu_print_prompt` (void)
Print a standard prompt for keyboard input ">".
- uint8_t `menu_retrieve_keypress` (uint32_t nUartAddress)
Get a single keypress via Hyperterminal.
- void `menu_print_main_menu` (void)
Print the main menu listing choice of module to test.
- void `menu_print_channel_menu` (void)
- void `menu_print_sample_rate_menu` (void)
Print a menu listing sample speed choices.
- void `menu_print_sample_size_menu` (void)
Print a menu listing sample size choices.

5.7.1 Detailed Description

Project: MAXREFDES61#
 Filename: menu.h
 Description: This module contains all the functions used to generate the menus and menu options used to run the MAXREFDES61 example firmware.

Revision History:

06-13-14 Rev 01.00 GL Initial Release

This code follows the following naming conventions:

char chPmodValue

char (array) sPmodString[16]

float fPmodValue

int nPmodValue

int (array) anPmodValue[16]

```
uint16_t uPmodValue
uint16_t (array) auPmodValue[16]
uint8_t uchPmodValue
uint8_t (array) auchPmodBuffer[16]
unsigned int unPmodValue
int * punPmodValue
Definition in file menu.h.
```

5.7.2 Macro Definition Documentation

5.7.2.1 **#define BASE_FUNCTION_STATE 10**

```
// Menu state machine state
Definition at line 74 of file menu.h.
```

5.7.2.2 **#define KEYPRESS_ARROW_DOWN 241**

Assign up-arrow an extended ascii code which won't be used elsewhere.
Definition at line 77 of file menu.h.

5.7.2.3 **#define KEYPRESS_ARROW_LEFT 242**

Assign up-arrow an extended ascii code which won't be used elsewhere.
Definition at line 78 of file menu.h.

5.7.2.4 **#define KEYPRESS_ARROW_RIGHT 243**

Assign up-arrow an extended ascii code which won't be used elsewhere.
Definition at line 79 of file menu.h.

5.7.2.5 **#define KEYPRESS_ARROW_UP 240**

Assign up-arrow an extended ascii code which won't be used elsewhere.
Definition at line 76 of file menu.h.

5.7.2.6 **#define KEYPRESS_END 244**

Assign up-arrow an extended ascii code which won't be used elsewhere.
Definition at line 80 of file menu.h.

5.7.2.7 **#define MAIN_MENU 0**

```
// Menu state machine state
```


Definition at line 67 of file menu.h.

5.7.2.8 **#define MENU_CHANNEL 6**

Definition at line 73 of file menu.h.

5.7.2.9 **#define MENU_SAMPLE_SIZE 3**

// Menu state machine state

Definition at line 70 of file menu.h.

5.7.2.10 **#define MENU_SAMPLE_SPEED 2**

// Menu state machine state

Definition at line 69 of file menu.h.

5.7.2.11 **#define START_BLOCK_SAMPLING 5**

// Menu state machine state

Definition at line 72 of file menu.h.

5.7.2.12 **#define START_CONTINUOUS_SAMPLING 4**

// Menu state machine state

Definition at line 71 of file menu.h.

5.7.2.13 **#define WAIT_KEYPRESS 1**

// Menu state machine state

Definition at line 68 of file menu.h.

5.7.3 Function Documentation

5.7.3.1 **void menu_cls (void)**

Function to clear the screen via Hyperterminal.

Parameters

<i>None</i>	
-------------	--

Return values

<i>None</i>	
-------------	--

Definition at line 64 of file menu.c.

5.7.3.2 void menu_print_channel_menu (void)

Definition at line 220 of file menu.c.

5.7.3.3 void menu_print_main_menu (void)

Print the main menu listing choice of module to test.

Details.

Return values

<i>None</i>	
-------------	--

Definition at line 203 of file menu.c.

5.7.3.4 void menu_print_maxim_banner (void)

Print standard Maxim banner at top of Hyperterminal screen.

Parameters

<i>None</i>	
-------------	--

Return values

<i>None</i>	
-------------	--

Definition at line 78 of file menu.c.

5.7.3.5 void menu_print_maxim_banner_big (void)

Print large Maxim banner at top of Hyperterminal screen.

Parameters

<i>None</i>	
-------------	--

Return values

<i>None</i>	
-------------	--

Definition at line 102 of file menu.c.

5.7.3.6 void menu_print_prompt (void)

Print a standard prompt for keyboard input " > ".

Parameters

<i>None</i>	
-------------	--

Return values

<i>None</i>	
-------------	--

Definition at line 135 of file menu.c.

5.7.3.7 void menu_print_sample_rate_menu (void)

Print a menu listing sample speed choices.

Details**Return values**

<i>None</i>	
-------------	--

Definition at line 233 of file menu.c.

5.7.3.8 void menu_print_sample_size_menu (void)

Print a menu listing sample size choices.

Details**Return values**

<i>None</i>	
-------------	--

Definition at line 252 of file menu.c.

5.7.3.9 uint8_t menu_retrieve_keypress (uint32_t nUartAddress)

Get a single keypress via Hyperterminal.

Details

Returns ascii character corresponding to keypress

Parameters

<i>in</i>	<i>nUartAddress</i>	- address of the UART peripheral in MicroBlaze memory map
-----------	---------------------	---

Return values

<i>Character, partially</i>	decoded.
-----------------------------	----------

Definition at line 162 of file menu.c.

5.8 src/retarget.c File Reference

```
#include <stdio.h>
#include <rt_misc.h>
#include "stm32f4xx_hal.h"
#include "usart.h"
```

Data Structures

- struct [__FILE](#)

Functions

- int [sendchar](#) (int c)
- int [getkey](#) (void)
- int [fputc](#) (int c, FILE *f)
- int [fgetc](#) (FILE *f)
- int [ferror](#) (FILE *f)
- void [_ttywrch](#) (int c)
- void [_sys_exit](#) (int return_code)

Variables

- FILE [__stdout](#)
- FILE [__stdin](#)

5.8.1 Function Documentation

5.8.1.1 void _sys_exit (int *return_code*)

Definition at line 60 of file retarget.c.

5.8.1.2 void _ttywrch (int *c*)

Definition at line 53 of file retarget.c.

5.8.1.3 int ferror (FILE * *f*)

Definition at line 47 of file retarget.c.

5.8.1.4 int fgetc (FILE * f)

Definition at line 39 of file retarget.c.

5.8.1.5 int fputc (int c, FILE * f)

Definition at line 33 of file retarget.c.

5.8.1.6 int getkey (void)**5.8.1.7 int sendchar (int c)****5.8.2 Variable Documentation****5.8.2.1 FILE __stdin**

Definition at line 30 of file retarget.c.

5.8.2.2 FILE __stdout

Definition at line 29 of file retarget.c.

5.9 src/spi.c File Reference

```
#include "spi.h"
#include "gpio.h"
```

Functions

- void [MX_SPI1_Init](#) (void)
- void [MX_SPI2_Init](#) (void)
- void [HAL_SPI_MspInit](#) (SPI_HandleTypeDef *hspi)
- void [HAL_SPI_MspDeInit](#) (SPI_HandleTypeDef *hspi)

Variables

- SPI_HandleTypeDef [hspi1](#)
- SPI_HandleTypeDef [hspi2](#)

5.9.1 Function Documentation**5.9.1.1 void HAL_SPI_MspDeInit (SPI_HandleTypeDef * hspi)**

SPI1 GPIO Configuration PA5 —> SPI1_SCK PA6 —> SPI1_MISO PA7 —> SPI1_MOSI

SPI2 GPIO Configuration PC2 ----> SPI2_MISO PC3 ----> SPI2_MOSI PB10 ----> SPI2_SCK PB12 ----> SPI2-_NSS

Definition at line 135 of file spi.c.

5.9.1.2 void HAL_SPI_MspInit (SPI_HandleTypeDef * hspi)

SPI1 GPIO Configuration PA5 ----> SPI1_SCK PA6 ----> SPI1_MISO PA7 ----> SPI1_MOSI

SPI2 GPIO Configuration PC2 ----> SPI2_MISO PC3 ----> SPI2_MOSI PB10 ----> SPI2_SCK PB12 ----> SPI2-_NSS

Definition at line 85 of file spi.c.

5.9.1.3 void MX_SPI1_Init (void)

Definition at line 49 of file spi.c.

5.9.1.4 void MX_SPI2_Init (void)

Definition at line 67 of file spi.c.

5.9.2 Variable Documentation

5.9.2.1 SPI_HandleTypeDef hspi1

File Name : [SPI.c](#) Date : 06/06/2014 14:31:14 Description : This file provides code for the configuration of the SPI instances.

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Definition at line 45 of file spi.c.

5.9.2.2 SPI_HandleTypeDef hspi2

Definition at line 46 of file spi.c.

5.10 src/stm32f4xx_it.c File Reference

Interrupt Service Routines.

```
#include "stm32f4xx_hal.h"
#include "stm32f4xx.h"
#include "stm32f4xx_it.h"
```

Functions

- void [SysTick_Handler](#) (void)
This function handles System tick timer.
- void [TIM1_UP_TIM10_IRQHandler](#) (void)
This function handles TIM1 Update interrupt and TIM10 global interrupt.

Variables

- TIM_HandleTypeDef [htim1](#)

5.10.1 Detailed Description

Interrupt Service Routines.

Date

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Definition in file [stm32f4xx_it.c](#).

5.10.2 Function Documentation

5.10.2.1 void SysTick_Handler (void)

This function handles System tick timer.

Definition at line 50 of file stm32f4xx_it.c.

5.10.2.2 void TIM1_UP_TIM10_IRQHandler (void)

This function handles TIM1 Update interrupt and TIM10 global interrupt.

Definition at line 59 of file stm32f4xx_it.c.

5.10.3 Variable Documentation

5.10.3.1 TIM_HandleTypeDef htim1

File Name : [TIM.c](#) Date : 06/06/2014 14:31:14 Description : This file provides code for the configuration of the TIM instances.

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Definition at line 43 of file tim.c.

5.11 src/tim.c File Reference

```
#include "tim.h"
```

Functions

- void [MX_TIM1_Init](#) (void)
- void [HAL_TIM_Base_MspInit](#) (TIM_HandleTypeDef *htim_base)
- void [HAL_TIM_Base_MspDeInit](#) (TIM_HandleTypeDef *htim_base)

Variables

- TIM_HandleTypeDef [htim1](#)

5.11.1 Function Documentation

5.11.1.1 void HAL_TIM_Base_MspDeInit (TIM_HandleTypeDef * *htim_base*)

Definition at line 85 of file tim.c.

5.11.1.2 void HAL_TIM_Base_MspInit (TIM_HandleTypeDef * *htim_base*)

Definition at line 69 of file tim.c.

5.11.1.3 void MX_TIM1_Init (void)

Definition at line 46 of file tim.c.

5.11.2 Variable Documentation

5.11.2.1 TIM_HandleTypeDef htim1

File Name : [TIM.c](#) Date : 06/06/2014 14:31:14 Description : This file provides code for the configuration of the TIM instances.

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Definition at line 43 of file tim.c.

5.12 src/usart.c File Reference

```
#include "usart.h"
#include "gpio.h"
```

Functions

- void [MX_USART1_UART_Init](#) (void)
- void [HAL_UART_MspInit](#) (UART_HandleTypeDef *huart)
- void [HAL_UART_MspDeInit](#) (UART_HandleTypeDef *huart)

Variables

- UART_HandleTypeDef [huart1](#)

5.12.1 Function Documentation

5.12.1.1 void HAL_UART_MspDeInit (UART_HandleTypeDef * *huart*)

USART1 GPIO Configuration PA9 ----> USART1_TX PA10 ----> USART1_RX PA11 ----> USART1_CTS PA12
----> USART1_RTS

Definition at line 89 of file usart.c.

5.12.1.2 void HAL_UART_MspInit (UART_HandleTypeDef * *huart*)

USART1 GPIO Configuration PA9 ----> USART1_TX PA10 ----> USART1_RX PA11 ----> USART1_CTS PA12
----> USART1_RTS

Definition at line 64 of file usart.c.

5.12.1.3 void MX_USART1_UART_Init (void)

Definition at line 49 of file usart.c.

5.12.2 Variable Documentation

5.12.2.1 UART_HandleTypeDef huart1

File Name : [USART.c](#) Date : 06/06/2014 14:31:15 Description : This file provides code for the configuration of the USART instances.

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Definition at line 45 of file usart.c.

5.13 *src/utilities.c* File Reference

```
#include "utilities.h"
#include "stm32f4xx_hal.h"
```

Functions

- void [Delay](#) (uint32_t unDelay)

5.13.1 Detailed Description

```
Project: MAXREFDES61#
Filename: utilities.c
Description: This module contains a collection of general utility
             functions which are not specific to any particular
             module.
```

Revision History:

06-13-14 Rev 1.0 GL Initial Release

This code follows the following naming conventions.

```
char chPmodValue
char (array) sPmodString[16]
float fPmodValue
int nPmodValue
int (array) anPmodValue[16]
uint16_t uPmodValue
uint16_t (array) auPmodValue[16]
uint8_t uchPmodValue
uint8_t (array) auchPmodBuffer[16]
unsigned int unPmodValue
int * punPmodValue
Definition in file utilities.c.
```

5.13.2 Function Documentation

5.13.2.1 void Delay (uint32_t unDelay)

Definition at line 64 of file [utilities.c](#).

5.14 src/utilities.h File Reference

```
#include "stm32f4xx_hal.h"
```

Macros

- #define [ONE_SECOND](#) 12500000

Functions

- void [Delay](#) (uint32_t unDelay)

5.14.1 Detailed Description

```
Project: MAXREFDES61#
Filename: utilities.h
Description: This module contains a collection of general utility
             functions which are not specific to any particular
             module.
```

Revision History:

06-13-14 Rev 1.0 GL Initial Release

This code follows the following naming conventions.

char chPmodValue

char (array) sPmodString[16]

float fPmodValue

int nPmodValue

int (array) anPmodValue[16]

uint16_t uPmodValue

uint16_t (array) auPmodValue[16]

uint8_t uchPmodValue

uint8_t (array) auchPmodBuffer[16]

unsigned int unPmodValue

int * punPmodValue

Definition in file [utilities.h](#).

5.14.2 Macro Definition Documentation

5.14.2.1 #define ONE_SECOND 12500000

Definition at line 65 of file [utilities.h](#).

5.14.3 Function Documentation

5.14.3.1 void Delay (uint32_t *unDelay*)

Definition at line 64 of file [utilities.c](#).

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